```
Error in square(i) : object 'i' not found
[1] TRUE
```

A number of other assert functions are provided by the RUnit package. The *checkEqualsNumeric* function checks if both the compared elements are equal and numeric. Otherwise, it returns an error, as illustrated below:

```
> checkEqualsNumeric(square(2), 4)
[1] TRUE

> checkEqualsNumeric(square(2), 3)
Error in checkEqualsNumeric(square(2), 3) :
  Mean relative difference: 0.25

> checkEqualsNumeric(square(2), 'a')
Error in checkEqualsNumeric(square(2), "a") : Modes: numeric,
character
target is numeric, current is character
```

The *checkIdentical* function is used to verify if the observed and expected values are equal, and it also has an additional logging mechanism. In the following example, two vectors, 'u' and 'v', are compared with each other as well as a list 'l'. A log message is printed if the compared elements are not identical.

```
> u <- c(1, 2, 3, 4)
> v <- c(1, 2, 3, 4)
> l <- list(1, 2, 3, 4)

> checkIdentical(v, l, "The types are different")
Error in checkIdentical(v, l, "The types are different") :
FALSE
 The types are different

> checkIdentical(u, v, "The types are different")
[1] TRUE
```

The *checkTrue* function takes a Boolean expression and a message, and asserts to see if the expression is true. Otherwise, it prints the message as shown below:

```
> checkTrue(TRUE, "True")
[1] TRUE

> checkTrue(FALSE, "False")
Error in checkTrue(FALSE, "False") : Test not TRUE
False

> checkTrue(5 > 2, "True")
[1] TRUE

> checkTrue(5 < 2, "False")
Error in checkTrue(5 < 2, "False") : Test not TRUE
False
```

The RUnit package also provides a tool for code inspection to obtain a test coverage report. This is useful to know which lines of code have been executed during test runs, and what additional tests need to be written. A tracker needs to be initialised, and the *inspect()* method needs to be invoked on the code with arguments. For example, the following commands demonstrate the sequence to run the inspection tool for the square function.

```
> track <- tracker()
> track$init()
> x <- 1:3
> resultSquare <- inspect(square(x), track=track)
> resTrack <- track$getTrackInfo()
> printHTML.trackInfo(resTrack)
```

A *results/* folder is then generated with the following file contents:

```
$ ls reports/
con1.html  con1.png  index.html  result1.html
```

The *result1.html* file that provides the code coverage report is shown in Figure 1.

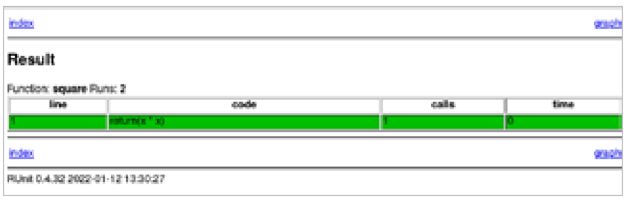

Figure 1: RUnit inspector report

For larger file modules, the connection graph between the functions and modules is also presented in the HTML output.

The *R CMD check* command can be used to build and also run the tests in an R source package. The *aucm* R package implements functions to identify linear and non-linear marker combinations. This maximises the area under the curve (AUC). It is released under the GPLv2 license and the tests are implemented using RUnit. A sample run of the tests is shown below:

```
$ R CMD check aucm
 ...
 * using R version 4.1.2 (2021-11-01)
 * using platform: x86_64-pc-linux-gnu (64-bit)
 * using session charset: UTF-8
```